

TEAM NUCLEAR FAMILY

Finding the best nuclear reactor sites in the United States



MEET THE TEAM



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AI IS STARVING FOR POWER

160%

data center electricity demand increase by 2030

3×

faster AI growth than grid expansion capacity

NOW

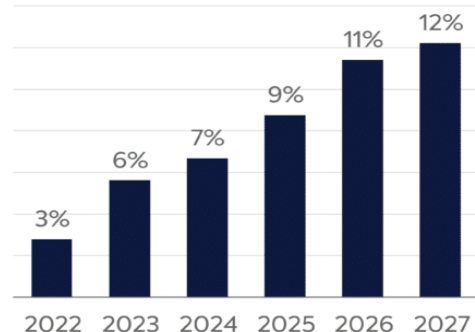
clean, reliable baseload power is needed — urgently

U.S. electricity demand is projected to surge **35–50% by 2040**, driven by data centers, manufacturing, and electrification.

via S&P Global Commodity Insights

FIGURE 2b — Demand projections

Figure 2b: AI LLM data center share of total commercial demand

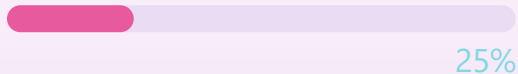


THE ONLY CLEAN ENERGY THAT CAN SCALE



SOLAR

RELIABILITY



STRENGTHS

- + Zero emissions
- + Low operating cost

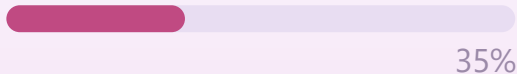
LIMITATIONS

- Intermittent (night/clouds)
- Land-intensive



WIND

RELIABILITY



STRENGTHS

- + Low emissions
- + Scalable offshore

LIMITATIONS

- Intermittent (calm days)
- Grid integration challenges



NUCLEAR

RELIABILITY



STRENGTHS

- + 24/7 baseload power
- + Tiny land footprint

LIMITATIONS

- High upfront capital
- Long permitting timeline

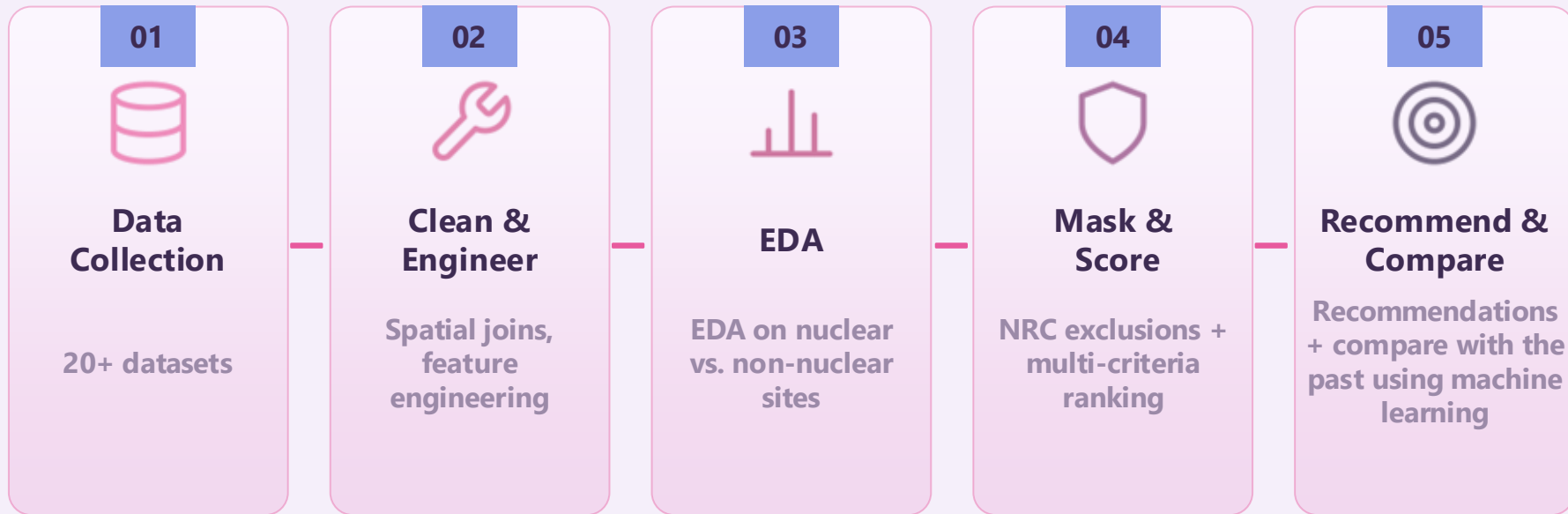
Nuclear is the only carbon-free source that runs 24/7 at scale.

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RESEARCH QUESTIONS & GOALS

- 1. Can we develop a modern framework to identify the best counties in the United States to have a nuclear reactor?*
- 2. Have the criteria for a "good" nuclear reactor site (involving both safety and demand) changed since the 1970-80s? If yes, how have they changed?*

A FIVE-STEP DATA-DRIVEN FRAMEWORK



Every step is guided by **Nuclear Regulatory Commission (NRC)** standards, expert consultation with **Professor Caleb Brooks** (Department of Nuclear, Plasma, and Radiological Engineering).

DATA INTRODUCTION

13

datasets integrated

3,235

US counties analyzed

28

features engineered

United States[®]
Census
Bureau

 **USGS**
science for a changing world

Find Energy 



 **U.S.NRC**

Hydro  **SHEDS**

OSTI.GOV



FEMA



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DATA CLEANING AND PREPROCESSING

1

SPATIAL JOIN TO COUNTY



Standardize every variable to county level

2

OVERLAY FOR PRECISION



Stacked layers (e.g., flood zones + counties) to measure only the overlapping areas within each county

3

FEATURE ENGINEERING



- number of nuclear plants
- % county in flood zones
- distance to rivers/lakes
- max transmission voltage

4

VARIABLE SELECTION + MERGE



- Kept 2–4 meaningful variables per dataset
- Merged everything into one county-level dataset

VARIABLES SELECTION

WHY?

- Ensure all variables in the model are relevant when evaluating a nuclear reactor site
- Remove redundant variables (measuring the same thing as others) to make sure weights are fairly distributed

HOW?

- **NRC grounding** – demand, seismic, population, grid access
- **Redundant/Overlapping variables** – e.g. *rivers* and *lakes* (*water access*)
- **County-level interpretability** – drop too-granular variables

VARIABLES THAT MOVE TOGETHER

HOUSING UNITS



**MEDIAN HOUSEHOLD
INCOME**

Both wealth signals — kept income.

**DIST TO
TRANSMISSION LINES**



DIST TO RIVERS

Lines follow rivers — kept rivers.

TOTAL RIVER MILES



RIVER COUNT

Both represent water access — kept total miles.

ENERGY CONSUMPTION

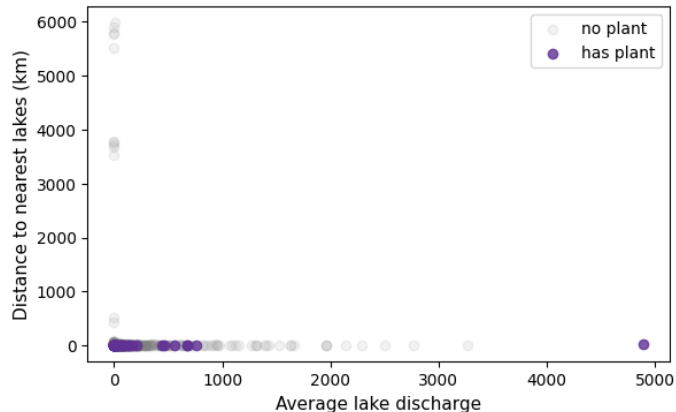


POPULATION

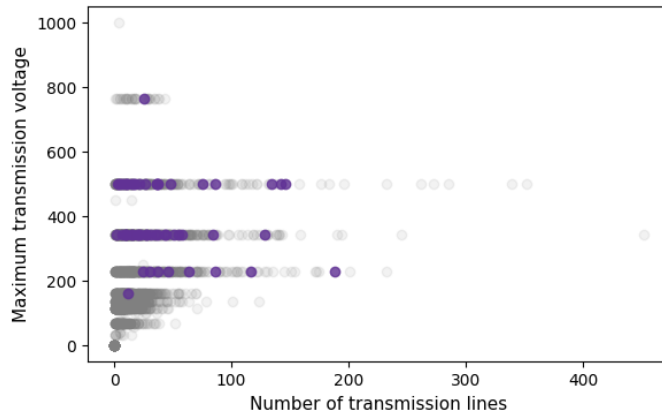
Both measure energy demand — kept energy.

EXPLORATORY DATA ANALYSIS

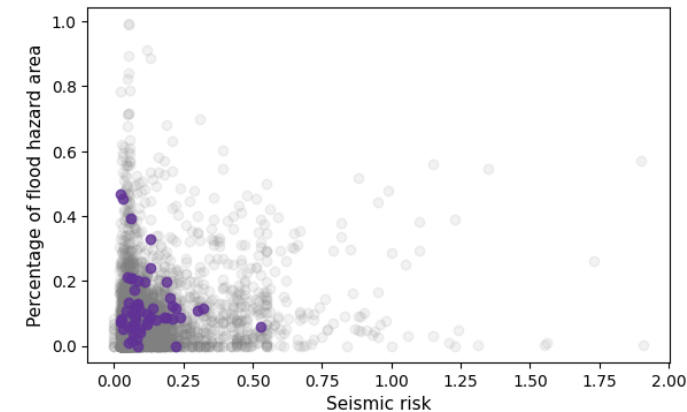
Lake Discharge vs. River Proximity



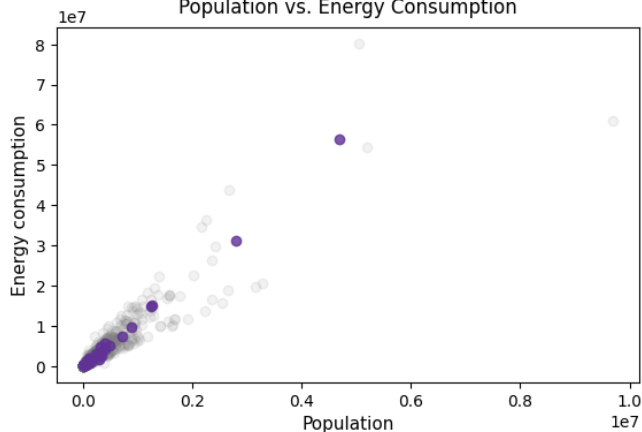
Transmission Line Count vs. Max Voltage



Seismic vs. Flood Risk



Population vs. Energy Consumption



Counties with nuclear plants:

- Closer to bodies of water
- Higher-voltage grid access
- Lower seismic and flood risks
- Lower population

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Where should America's next reactors be built?

Building a modern framework to site the next U.S.
reactor

MODELING: HARD EXCLUSIONS

SEISMIC RISK

Threshold: $\text{PGA} > 0.3\text{g}$

Source: USGS / NRC

FLOOD HAZARD

Threshold: $> 20\%$ area in SFHA

Source: FEMA / Exec. Order

MILITARY OVERLAP

Threshold: $> 10\%$ county area

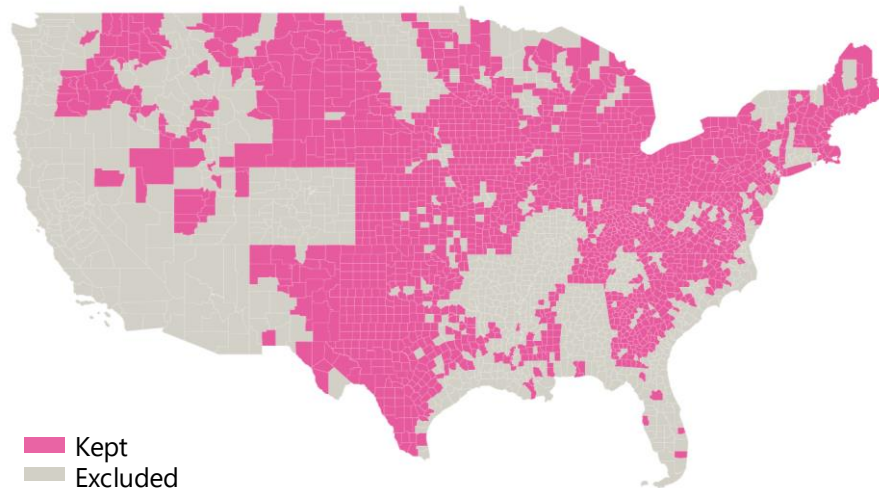
Source: Dept. of Defense

PROTECTED LAND

Threshold: $> 10\%$ county area

Source: NRC / NGNP

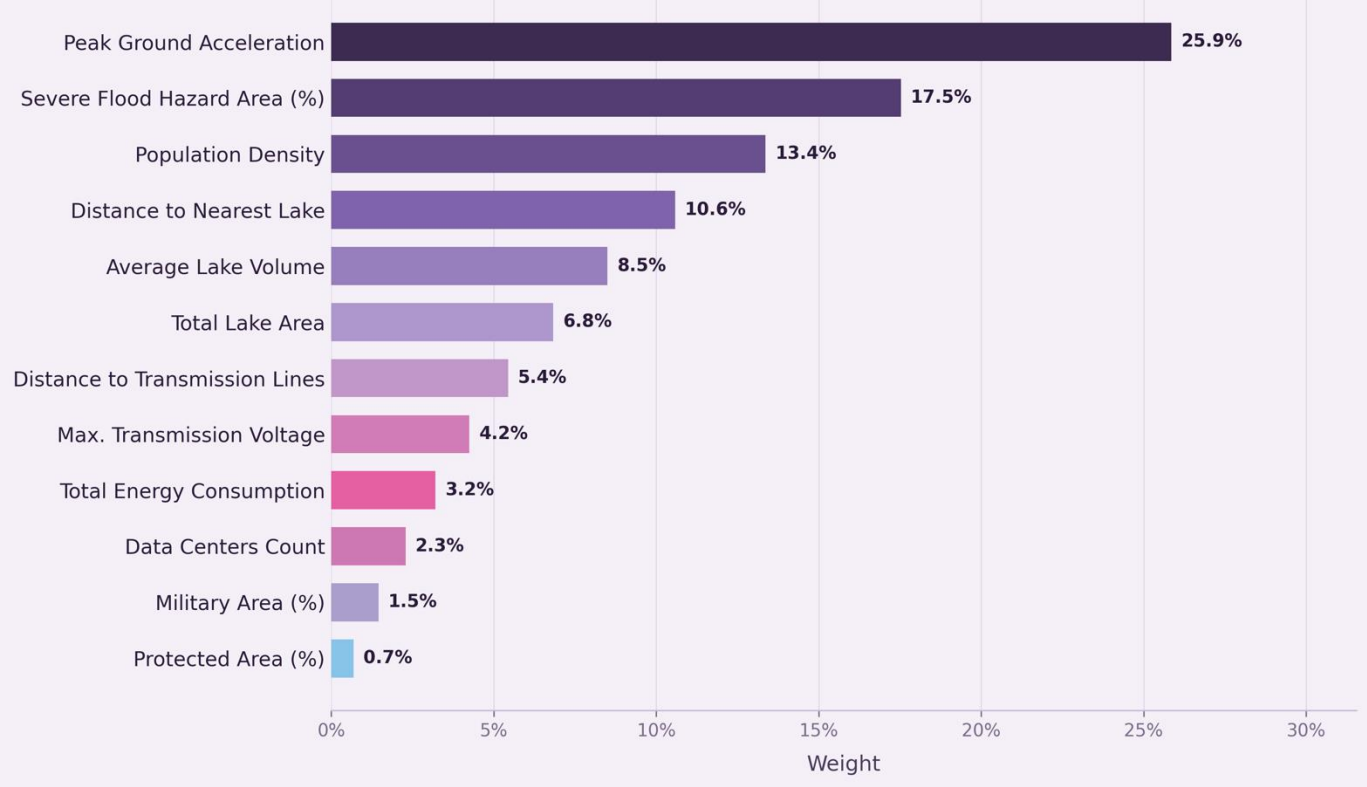
1,094
COUNTIES ELIMINATED



MULTI-CRITERIA DECISION ANALYSIS (MCDA)



MCDA FEATURES IMPORTANCE RANKING



SENSITIVITY ANALYSIS

Why: *Our weights might not be perfect, and ROC method may not capture the nuanced gap in the importance of different factors*

METHOD

- 1 PERTURB**
Randomly shift each variable's weight by $\pm 20\%$
- 2 SIMULATE**
Re-run the MCDA score 1,000 times
- 3 COMPARE**
Count how many of the top 20 counties stay on top

RESULT

18.8 / 20

counties stay in the top 20 on average

95% CI: [18.74, 18.85]

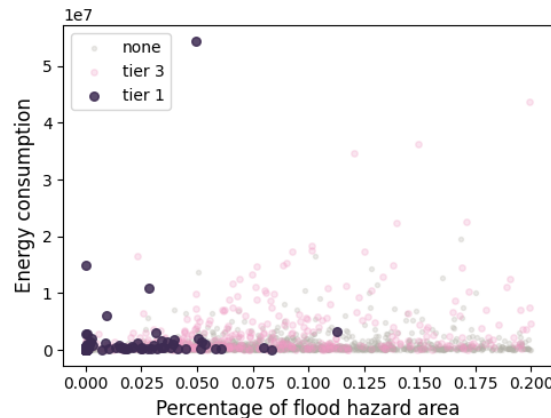
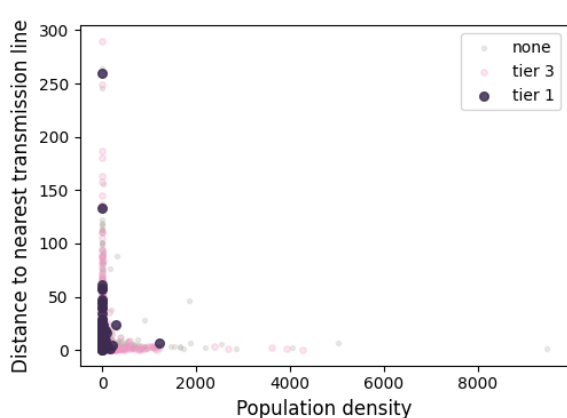
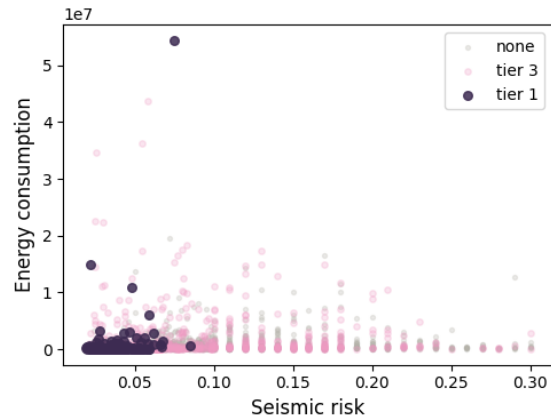
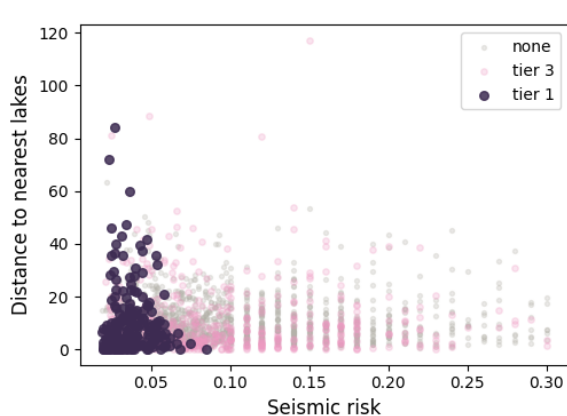
MULTI-OBJECTIVE OPTIMIZATION (MOO)

Pairwise objective views

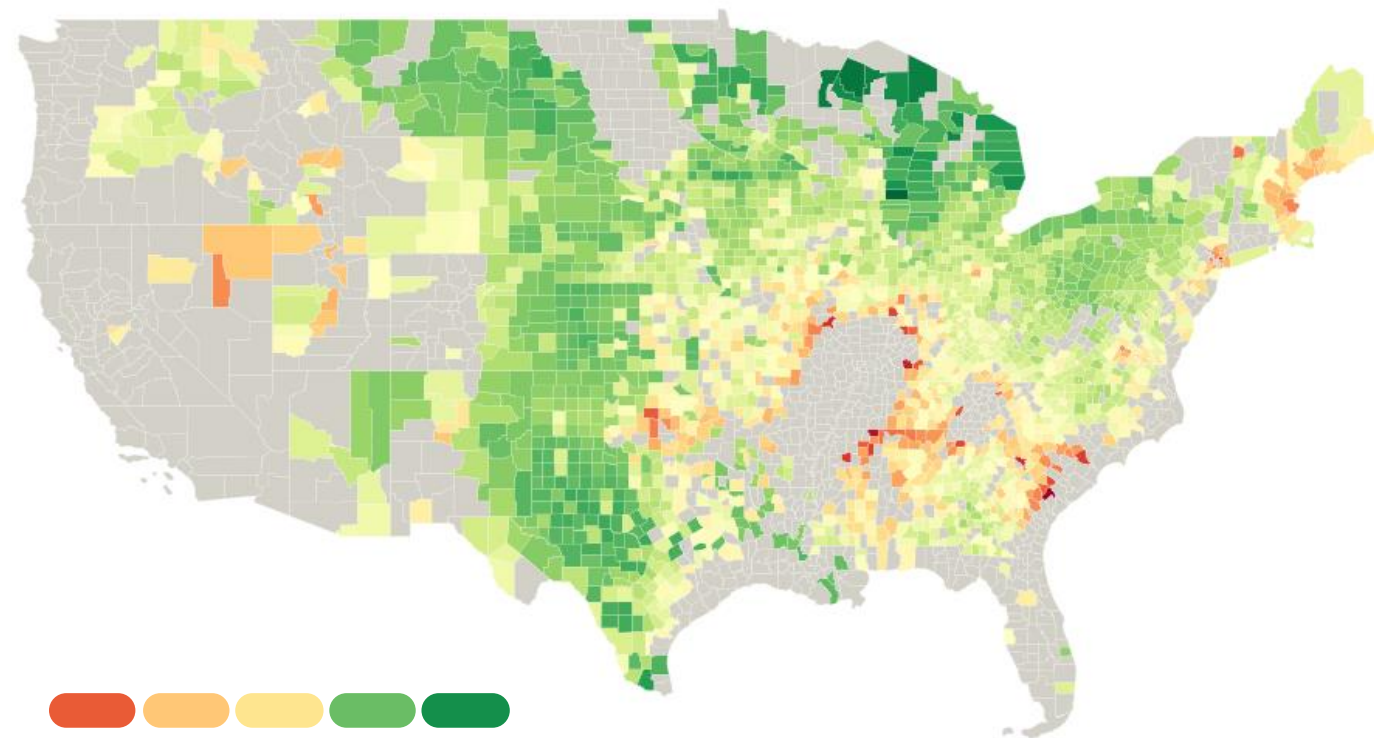
Why: Find the optimal nuclear counties for a nuclear reactor that are non-dominated in any aspects

What:

- 175 optimal counties
- Both Top 20 from MOO and MCDA have **18 counties** in common



RECOMMENDATIONS



Less Suitable

More Suitable

Top 5 Counties

Milwaukee Co., WI
0.811

Ontonagon Co., MI
0.794

Ashland Co., WI
0.790

Alger Co., MI
0.784

Houghton Co., MI
0.784

Back to the Past

Comparing our approach to historical siting decisions

Most past reactors were built in the 1970s–80s

WHAT DID THE PAST PRIORITIZE?

TRAINING DATA

2,141

counties
analyzed

41

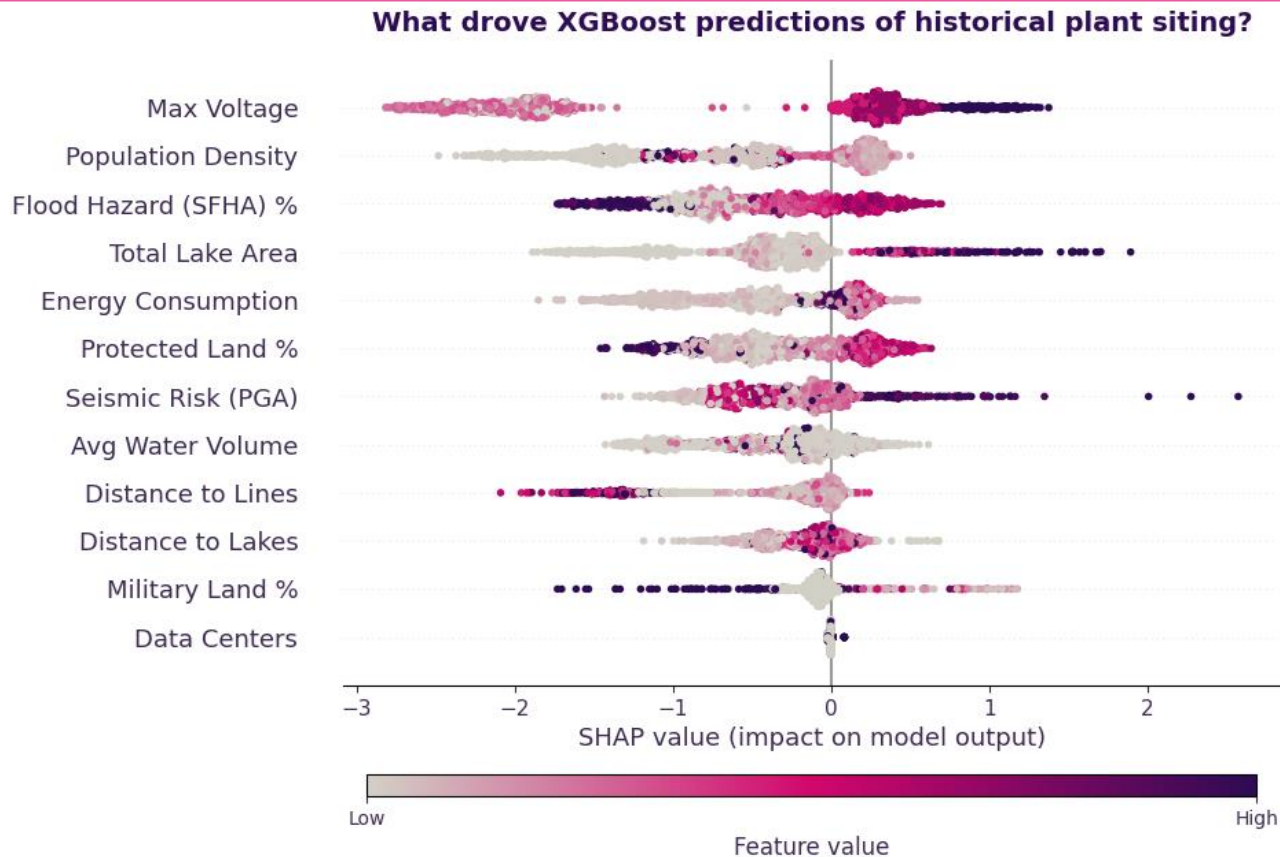
counties with
existing plants

- Our top counties do not contain existing plants
- Max voltage found to be the most important
- Found permutation importances as a sanity check

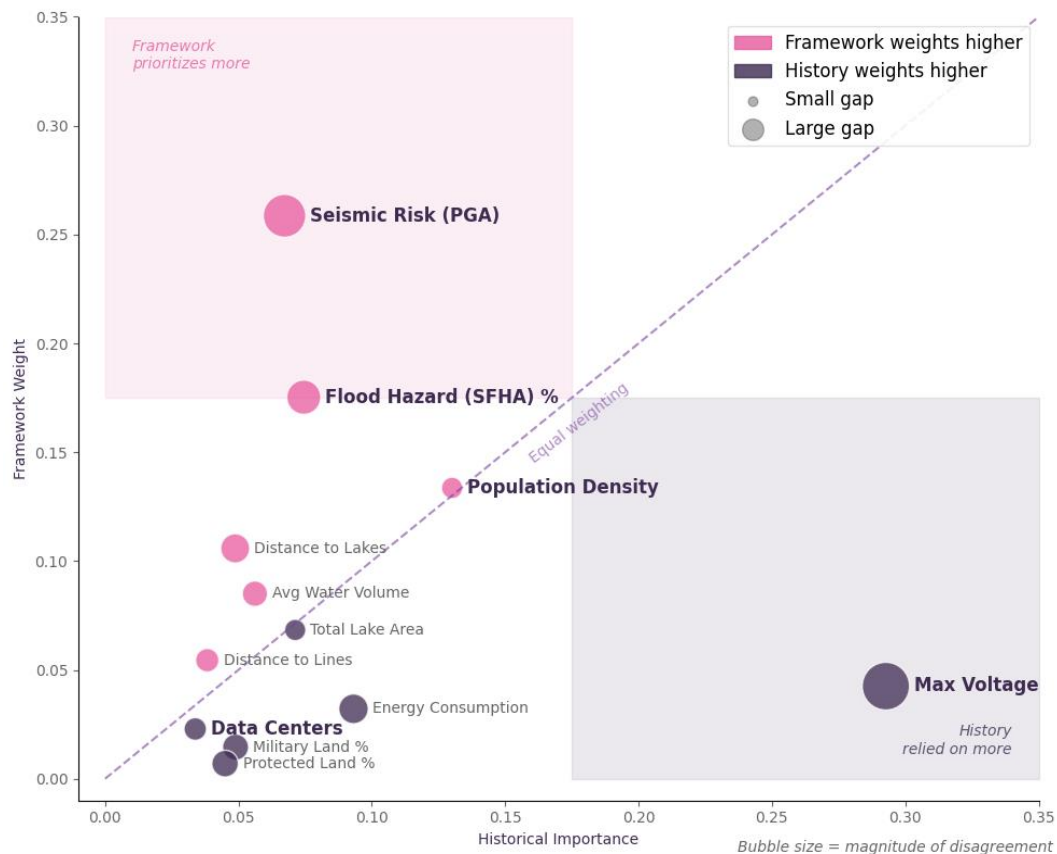
Best Performance for Counties with Existing Plants

Model	Test F1	Test Precision	Test Recall
Logistic Regression	0.114	0.062	0.750
Decision Tree	0.129	0.087	0.250
XGBoost	0.180	0.120	0.375

WHAT DID THE PAST PRIORITIZE?



COMPARISON WITH THE PAST



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ROADBLOCKS AND FUTURE WORK

ROADBLOCKS

- Data heterogeneity across sources
- Modern vs. legacy siting criteria
- Heavy class imbalance (41 of 2,141 counties)
- Limited publicly available cost data

FUTURE WORK

- Validate against OR-SAGE, the DOE's established site-screening tool, recommended by Prof. Brooks as a credibility benchmark
- Add economic & cost layer to scoring
- County-level public sentiment analysis



QUESTIONS?